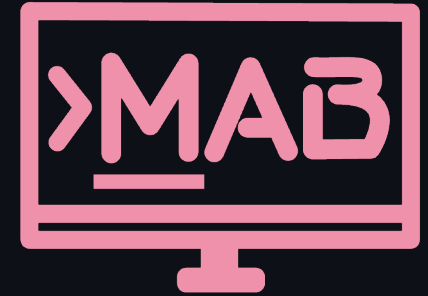


MICROBIAL EVOLUTION & VARIANT CALLING (Breseq)



Arkadiy I. Garber & Vaughn S. Cooper

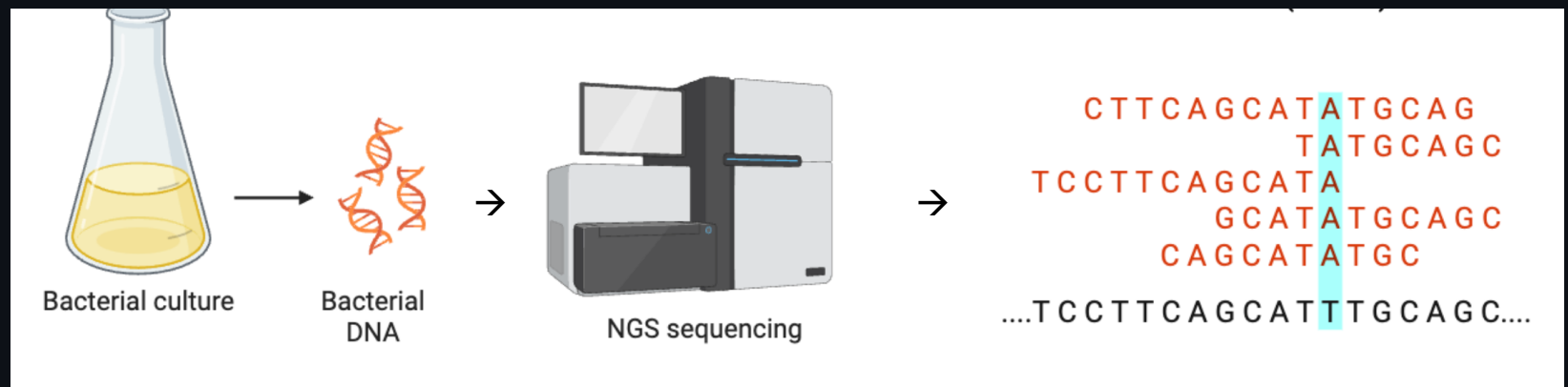
Middle Author Bioinformatics

One Health Microbiome Symposium · May 14, 2026 · Penn State



University of Pittsburgh

Microbial Variant Calling



Workshop Overview

Hands-On Practice

Tour of the MAB breseq web app
Upload reads / select reference
Initiate variant calling pipeline



Theory & Concepts

Experimental evolution & its goals
NGS sequencing and read alignment
Variant types in evolved microbes
Introduction to breseq
Reference genome selection
Reading & interpreting breseq output

Hands-On Practice

Explore the interactive report
Interpret mutations biologically
Q&A and open exploration



Navigate to breseq.midauthorbio.com



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Create an Account

Name:

Email:

Institution:

Lab or Division (optional):

Do you need bioinformatics support? (optional)

☒ No ☐ Yes

Submit Registration

Clear

SIGN IN

Submit some data!

Analysis Name:

Upload Forward Reads:

Choose File No file chosen

Upload Reverse Reads (optional):

Choose File No file chosen

Choose Reference Type:

Use Assembly Accession

Upload Reference Files

Reference GenBank File (.gbk):

Choose File No file chosen

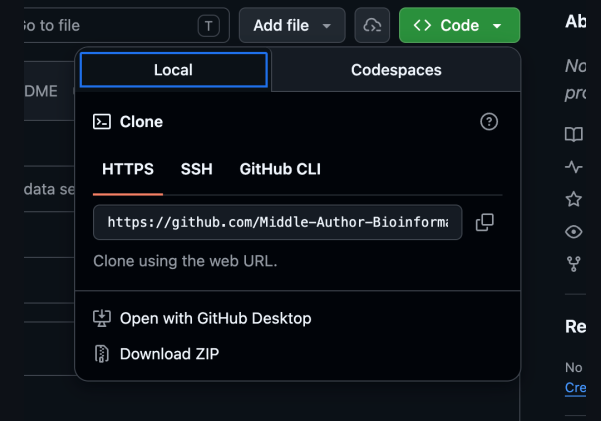
What kind of sample are you submitting?

Clonal

Submit Clear

github.com/Middle-Author-Bioinformatics/OneHealth2026

Middle-Author-Bioinformatics	Delete .DS_Store	7c14b00 · now	12 Commits
A_2019_1_fwdpaired.fastq.gz	new files	now	
A_2019_1_revpaired.fastq.gz	new files	now	
Pseudomonas_fluorescens_SBW25_116.gbk	new files	now	
README.md	Add sample data section to README	2 weeks ago	



<https://www.ncbi.nlm.nih.gov/genbank/>

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Why Sequence DNA?

Whole-genome sequencing as an evolutionary tool

< \$100

Cost per genome
(Illumina WGS)

99.9%

Accuracy of
short-read SNP calls

100s

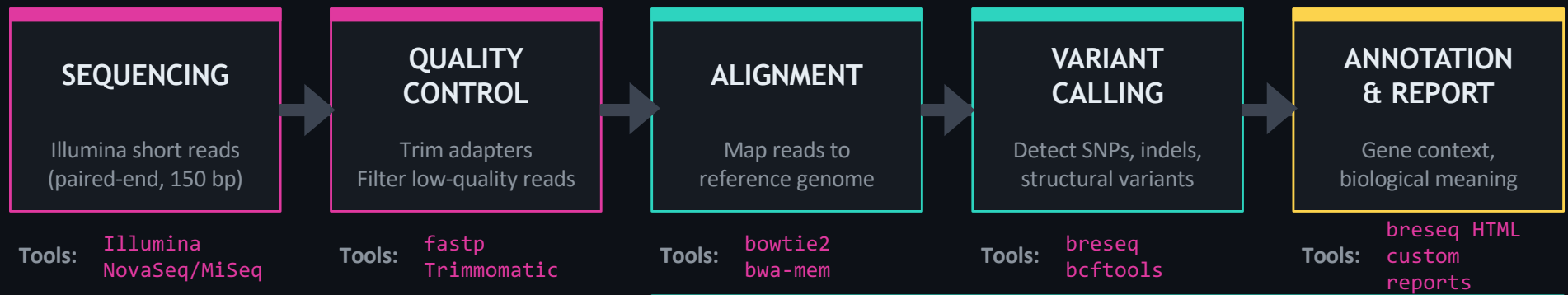
Mutations detectable
across whole genome

What It Enables

- Experimental Evolution
- In-Host Pathogen Evolution
- AMR Surveillance
- Transmission & Outbreak Tracing
- Genetic Construct Verification
- Symbiont / Mutualist Adaptation

From Reads to Mutations

The complete variant-calling workflow



breseq handles alignment + variant calling + annotation in one command

Short-Read Sequencing Basics

Illumina sequencing — what you get and what it means

Illumina Technology

Read length

100–300 bp (paired-end 2×150 bp typical)

Paired-end

Two reads from both ends of a fragment ~300–500 bp

Coverage depth

Number of reads overlapping each position

FASTQ format

4 lines per read: ID, sequence, "+", quality scores

FASTA format

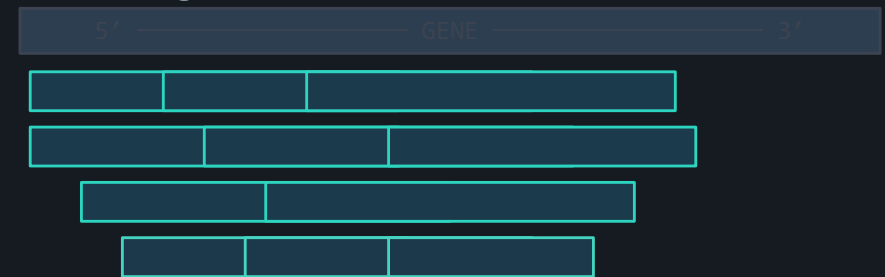
2 lines per sequence: ID, sequence

GenBank format

Sequences annotated and stored in in a vertical record

Coverage = Depth of Evidence

Reference genome



Reads pile up at each position

$$\text{Coverage} = \frac{(\text{read count} \times \text{read length})}{\text{genome size}}$$

Recommend $\geq 30\times$ for clonal · $\geq 100\times$ for polymorphic populations

Mutation Types in Evolved Microbes

SNP / SNV

A → G

Single-nucleotide substitutions. Most common. Can be synonymous (silent) or nonsynonymous (amino acid change).

Small Indel

+ATCG / -2 bp

Insertions or deletions of 1–50 bp. Often frameshifting in coding sequences → loss-of-function.

IS Element Insertion

IS1 → gene

Transposable insertion sequences jump into genes or promoters. Classic mechanism in *E. coli* evolution. Breseq detects via split reads at junction boundaries.

Large Deletion

-5 kb region

Loss of entire operons or metabolic pathways. Detected via zero-coverage windows (missing coverage). Often adaptive under nutrient limitation.

Amplification / CNV

×3–20 copies

Gene copy number increases detected by elevated coverage depth in a region. Often seen as a first response to selection pressure.

Mobile Element

+mob/phage

Prophage integrations, plasmid rearrangements, and mobile genetic element activity. Detected via junction candidates and coverage breaks.

Introducing Breseq

Best-in-class pipeline for microbial variant calling

Deatherage and Barrick, 2014 | doi: 10.1007/978-1-4939-0554-6_12

Generic Variant Callers (GATK, bcftools)

- Designed for human resequencing
- Population genetics / germline variants
- Poor IS element / junction detection
- Complex multi-step workflows
- No HTML report output

vs.

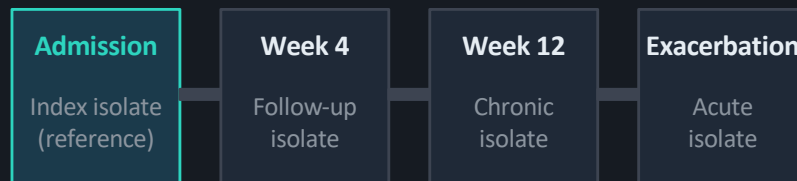
breseq

- Built for haploid microbial genomes
- Detects SNPs, indels, AND structural variants
- Identifies novel sequence junctions
- One command → full annotated HTML report
- Polymorphism mode for evolving populations

Tracking Evolution

Pathogens adapting during infection · Symbionts adapting during host association

Pathogen In-Host Evolution



Classic examples

P. aeruginosa in CF lungs; *Staph aureus* in prosthetic joint infection

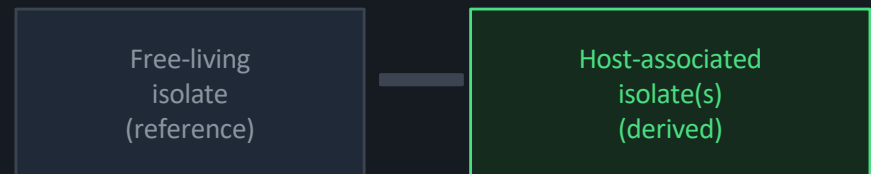
What breseq finds

Mutations in virulence regulators, biofilm genes, antibiotic efflux pumps, LPS modification loci

Key design note

Sequence the admission isolate as reference — instead of using a type strain — to eliminate pre-existing SNPs

Symbiont & Mutualist Adaptation



Classic examples

P. syringae on plants ; insect (mealybugs, aphids, etc.) endosymbionts

What breseq finds

Reductive evolution (deletions of free-living genes), IS element reshuffling, host-sensing regulator mutations

Why not a type strain?

Closely related environmental strains carry background SNPs that swamp the signal — use your own starting isolate

AMR Surveillance & Genetic Construct Verification

Clinical microbiology and genetic engineering QC

AMR Emergence & Transmission

De novo emergence

- Susceptible → resistant isolate during treatment
- Finds causal mutations: *rpoB*, *gyrA*, *bla*, etc.

Transmission tracing

- Shared SNP signature links patient-to-patient spread
- Distinguishes de novo from acquired resistance

Serial isolate surveillance

- Track resistance evolution across an ICU cohort
- Detect compensatory mutations restoring fitness

Genetic Construct Verification

Expected

Intended edit present
at correct locus

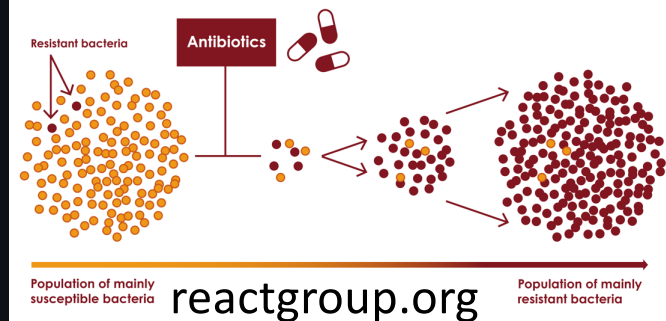
Off-target?

Additional SNPs or
IS insertions flagged

Why not just Sanger?

Sanger checks one locus; breseq covers the whole genome

Natural selection of resistant bacteria



How Breseq Works

Internal pipeline logic

Read Preprocessing

Trim adapters, filter quality (optional fastp/Trimmomatic)

Alignment (bowtie2)

Map reads to reference; identify unaligned portions

Junction Detection

Unaligned read ends → candidate new sequence junctions (IS insertions, deletions, inversions)

Coverage Analysis

Identify regions of missing, elevated, or uneven coverage (deletions, amplifications)

Variant Calling

Bayesian model scores each base position for SNPs and indels, with Phred-scaled evidence

Annotation & Output

Assign mutations to genes, annotate codon changes, write HTML + GD format output

```
$ sample breseq command
```

```
breseq \  
  -r reference.gbk \  
  reads_R1.fastq.gz \  
  reads_R2.fastq.gz
```

Output Files

index.html Interactive mutation report

output.gd Genome diff — machine-readable

output.vcf VCF for downstream analysis

evidence/ Read pileup HTML for each call

log.txt Full pipeline run log

Reference Genome Selection

 The reference genome you choose determines what variants breseq can and cannot detect. A mismatched reference introduces false positives and masks real mutations.

BEST CHOICE

Ancestral Strain Genome

The actual starting strain of your evolution experiment, sequenced before selection began. Eliminates all standing variation and gives highest sensitivity.

- Precisely matches your experimental organism
- Requires prior sequencing / assembly effort

GOOD CHOICE

NCBI Reference Accession

A finished, annotated genome for the same species / strain from NCBI. Best when no ancestral sequence is available.

- Widely available, high-quality annotation
- Small genomic differences → background noise
- Use when ancestral sequence unavailable

USE WITH CARE

Draft / Short-Read Assembly

A de novo assembly from short reads. Common gaps and fragmented contigs reduce alignment accuracy and create false junction signals.

- Assembly gaps → spurious variant calls
- Missing regions = invisible mutations

Reading Breseq Output

Evidence types

RA

Read Alignment

JC

New Junction

MC

Missing Coverage

Predicted mutations

SNP

Single-nucleotide
polymorphism

DEL

Deletion

INS

Insertion

MOB

Mobile element
insertion

AMP

Amplification / CNV

INS

Insertion

Note: Frequency < 100% indicates a polymorphic population — requires breseq polymorphism mode (-p flag)

Experimental Design Considerations

Before you sequence: decisions that shape your data

Coverage Depth

- $\geq 30\times$ minimum for clonal endpoint sequencing
- $\geq 100\times$ recommended for polymorphism detection
- Lower coverage $\rightarrow$ missed low-frequency variants

Ancestral Controls

- Always sequence the ancestral starting strain and use long reads!
- Establishes reference for variant calling
- Distinguishes standing vs. de novo variation

Biological Replication

- Minimum 3–6 independent evolved lineages
- Separates adaptive from neutral mutations
- More replicates = better statistical power

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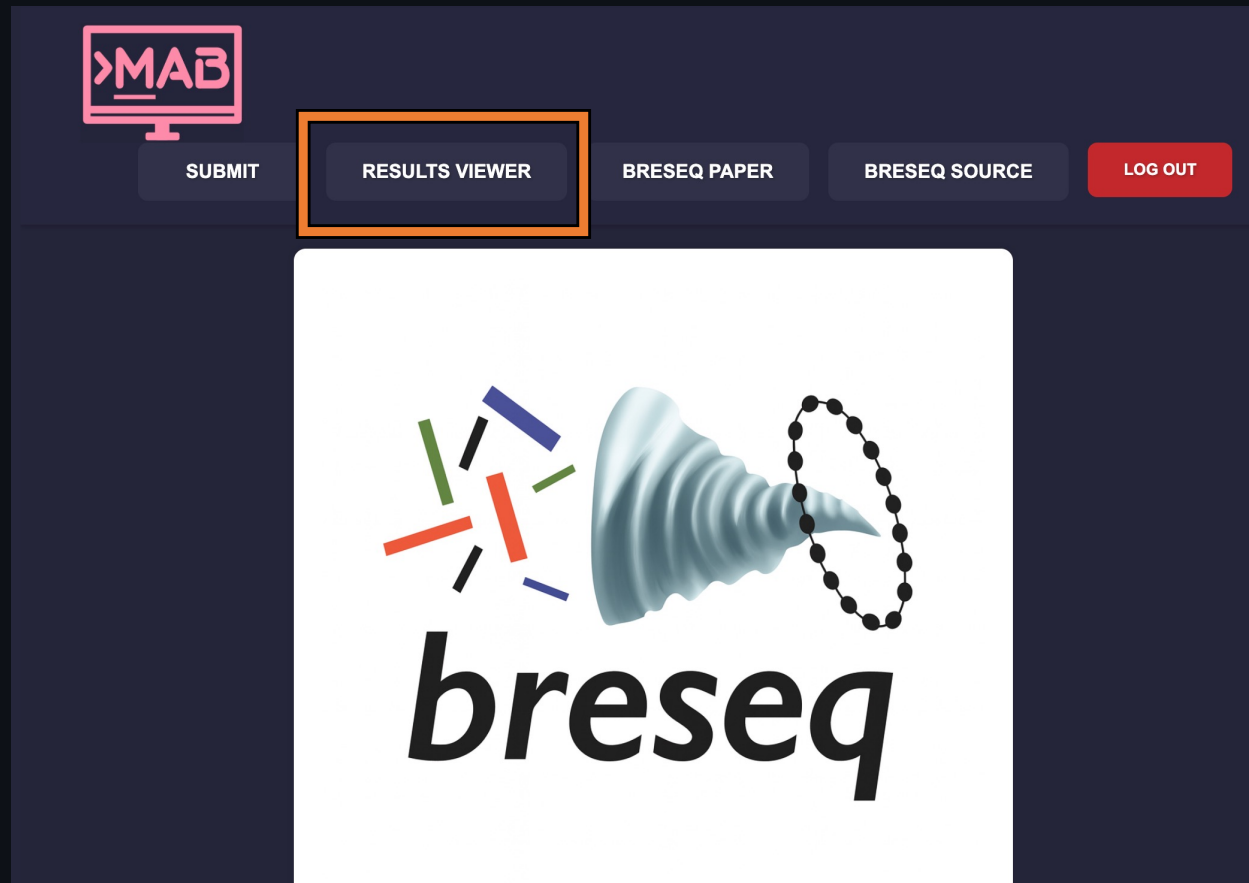
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Breseq Analyses

AMM106A_sGP2zyyLsO

P10_OyecZ1rqlW

P11_cBY0Auj11y

P2_qpGBYhTrwr

P3_EYcexFMRX3

P4_d9WMgYmwES

P5_qxliOCvSmm

P6_2jfnrPPCqY

P8_1yjUk7oEZ6



breseq version 0.37.0

[mutation predictions](#) | [marginal predictions](#) | [summary statistics](#) | [genome diff](#) | [command line log](#)

Predicted mutations

evidence	position	mutation	annotation	gene	description
RA	1,354,262	G→A	W54* (T G G→T A G)	<i>PFLU_1220</i> →	Putative chemotaxis-related protein
RA	1,359,378	C→T	Q185* (C A G→ T AG)	<i>PFLU_1224</i> →	Protein-glutamate methylesterase/protein-glutamine glutaminase (EC (EC
RA	2,479,701	T→A	L242Q (C I G→C A G)	<i>PFLU_2282</i> →	Putative ABC transport system, ATP-binding protein

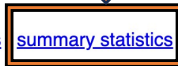
Unassigned new junction evidence

	seq id	position	reads (cov)	reads (cov)	score	skew	freq	annotation	gene	
+	2	OV986001 = 360368	20 (0.000)	13 (0.000)	7/184	NT	43.0%	intergenic (+29/-273)	<i>PFLU_0325/PFLU_0327</i>	Putative membrane protein/Imidazoleglycerol-phosphate dehydratase (IGPD) (EC
-	2	OV986001 360498 =	16 (0.000)					intergenic (+159/-143)	<i>PFLU_0325/PFLU_0327</i>	Putative membrane protein/Imidazoleglycerol-phosphate dehydratase (IGPD) (EC

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Sanity checks



Breseq Analyses

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P3_EYcexFMRX3

P4_d9WMgYmwES

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Summary statistics



Breseq Analyses

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P4_d9WMgYmwES

P5_qxliOCvSmm

P6_2jfnrPPCqY

P8_1yjUk7oEZ6

PRESET2_8LWrxpSO...

PresetTest_51t4HQucCf

REF2_4VEJQV7Msn



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Read File Information

	read file	reads	bases	passed filters	average	longest	mapped
errors	P3_S10_L001_R1_001	727,982	73,526,182	100.0%	101.0 bases	101 bases	95.9%
errors	P3_S10_L001_R2_001	722,809	73,003,709	99.3%	101.0 bases	101 bases	94.2%
	total	1,450,791	146,529,891	99.6%	101.0 bases	101 bases	95.0%

Reference Sequence Information

		seq id	length	fit mean	fit dispersion	% mapped reads	description
coverage	distribution	NC_012660	6,722,539	20.3	2.5	100.0%	Pseudomonas fluorescens SBW25, complete sequence.
		total	6,722,539			100.0%	

fit dispersion is the ratio of the variance to the mean for the negative binomial fit. It is =1 for Poisson and >1 for over-dispersed data.

Mutation predictions

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Breseq Analyses

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P2_qpGBYhTrwr

P3_EYcexFMRX3

P4_d9WMgYmwES

P5_qxliOCvSmm

P6_2jfnrPPCqY

P8_1yjUk7oEZ6



breseq version 0.37.0

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Unassigned new junction evidence

	seq id	position	reads (cov)	reads (cov)	score	skew	freq	annotation	gene	
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- 2	OV986001	360498 =	16 (0.000)					intergenic (+159/-143)	PFLU_0325/PFLU_0327	Putative membrane protein/Imidazoleglycerol-phosphate dehydratase (IGPD) (EC

The area in the red box describes your predicted mutation(s).

Predicted mutations

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- **Column 1:** the evidence used by breseq to determine if a mutation was present
- **Column 2:** the genome position where the mutation was found
- **Column 3:** the change in the DNA sequence from the ancestor to the mutant
- **Column 4:** the amino acid and codon change from the ancestor to the mutant
- **Column 5:** a unique identifier for the gene that was mutated
- **Column 6:** a description of the gene's predicted function, if known

position
1,354,262

Where to locate the mutation on the *P. fluorescens* genome

mutation

G→A

This mutation was a single nucleotide change.

The *P. fluorescens* ancestor had a G, but the evolved mutant had a A.

annotation

W54* (TGG→TAG)

This single nucleotide change altered the codon sequence.

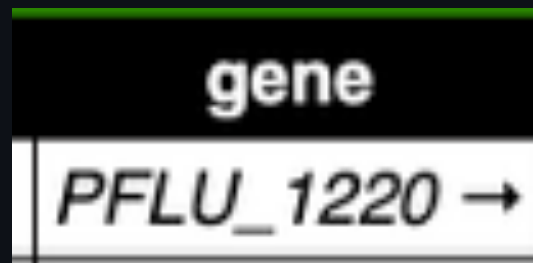
The *P. fluorescens* ancestor had TGG, but the evolved mutant had TAG.

annotation

W54* (TGG→TAG)

This change occurred at amino acid position 54 and changed the amino acid that will be present in the protein this gene encodes.

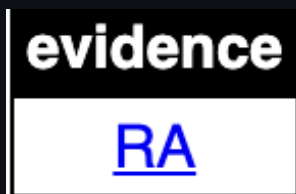
The *P. fluorescens* ancestor had Tryptophan (shorthand = W), but the evolved mutant had a stop codon (shorthand = *).



a unique identifier for the gene that was mutated

You can use this information to learn about your mutated gene.

A good resource for this is the Pseudomonas Genome Database (<http://www.pseudomonas.com/>).

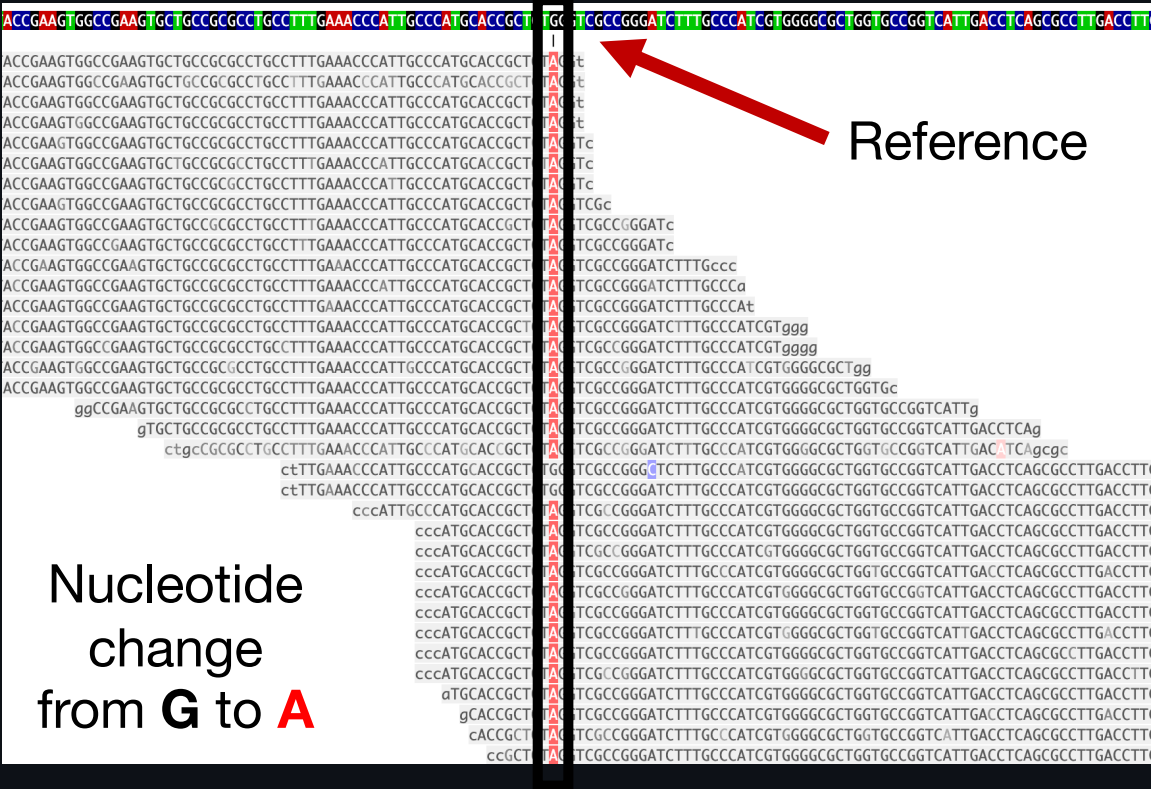


the evidence used by breseq to determine if a mutation was present

Clicking on the link in column 1 provides detailed information on the evidence breseq used to identify the mutation

Predicted mutations

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New junction evidence



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P11_cBY0Auj1y

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P5_qxliOCvSmm

P6_2jfnrPPCqY

P8_1yjUk7oEZ6



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Missing coverage evidence



breseq version 0.
mutation prediction

Predicted mutations

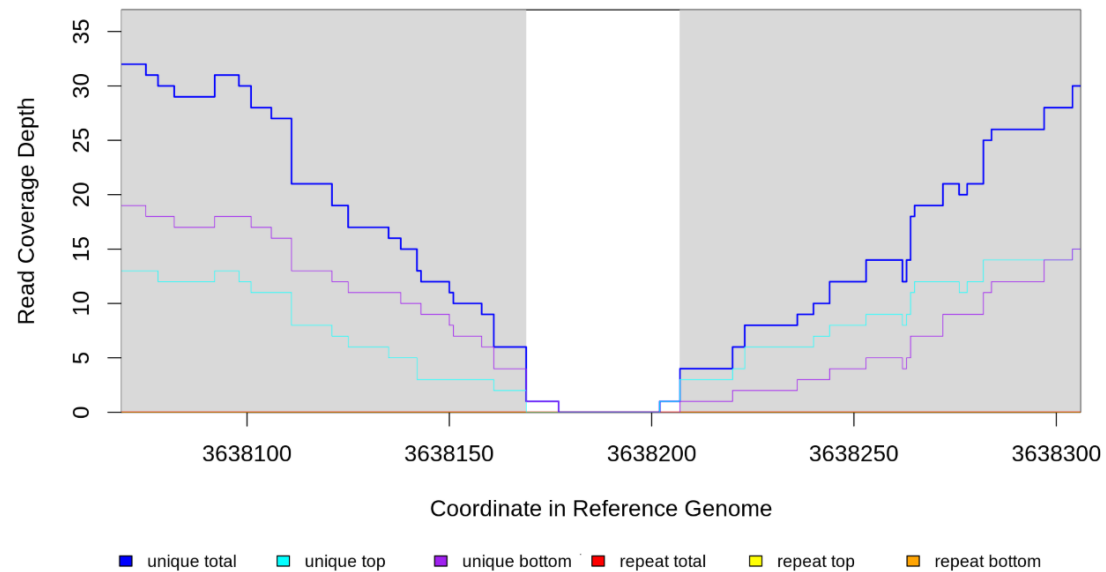
evidence	position	mutation
RA	45,881	+C
RA	985,333	+C
RA	1,359,168	C→
RA	3,447,984	(C)5

Assigned missing coverage

seq id	start
NC_012660	345139
NC_012660	363816
NC_012660	447994
NC_012660	447997
NC_012660	571735
NC_012660	571736

Missing coverage evidence...

seq id	start	end	size	←reads	reads→	gene	description
NC_012660	3638169	3638206	38	6 [1]	[1] 4	PFLU3288/PFLU3289	hypothetical protein/hypothetical protein



Marginal predictions

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AMM106A_sGP2zyLsO

P10_OyecZ1rqIW

P11_cBY0Aujl1y

P2_qpGBYhTnwr

P3_EYcexFMRX3

P4_d9WMgYmwES

P5_qxliOCvSmm

P6_2fhrPPCqY

P8_1yJk7oEZ6

PRESET2_8LWxpSO...

PresetTest_5114HQucCf

REF2_4VEJQV7Msn

REF_NJr5dJeHvt

REL606_5TlbpYQ2Mv

REL606_gSMVIVvWnj

Yan13_MR3XizNjI

Yan9_Z3FMIu6Wxw

submit_time_9hJ3JAKIT



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Marginal read alignment evidence (highest frequency 20 of 39 shown, sorted by frequency from high to low)

seq id	position	ref	new	freq	score (cons/poly)	reads	annotation	genes		
* NC_012660	598,561	0	C	A	60.0%	3.0 / 10.1	13	intergenic (+28/+81)	PFLU_RS02600/PFLU_RS02605	methyl-accepting chemotaxis protein/iron ABC transporter permease
* NC_012660	6,701,895	0	G	A	60.0%	-6.8 / 14.9	20	intergenic (-106/+18)	glmU/PFLU_RS30115	bifunctional UDP-N-acetylglucosamine diphosphorylase/glucosamine-1-phosphatase
* NC_012660	468,364	0	C	A	53.8%	8.0 / 16.3	13	intergenic (+98/+75)	PFLU_RS02065/PFLU_RS02070	LysR family transcriptional regulator/hypothetical protein
* NC_012660	4,077,157	0	G	A	53.8%	9.0 / 15.2	13	intergenic (+155/+89)	PFLU_RS17960/PFLU_RS17965	LLM class flavin-dependent oxidoreductase/class II aldolase/adducin family protein
* NC_012660	4,552,991	0	G	A	52.9%	1.9 / 14.2	17	intergenic (-171/-15)	PFLU_RS20110/PFLU_RS20115	Lrp/AsnC family transcriptional regulator/zinc-dependent alcohol dehydrogenase
* NC_012660	6,416,554	0	C	T	50.0%	-0.3 / 13.2	18	intergenic (-7/-364)	PFLU_RS28850/tauA	OprD family porin/taurine ABC transporter substrate-binding protein
* NC_012660	3,832,994	0	C	T	47.1%	-3.6 / 22.6	17	G147D (GCG→GAC)	PFLU_RS16905	DUF2252 family protein
* NC_012660	3,221,836	0	C	T	40.0%	13.0 / 10.5	20	A43V (GCG→GTG)	PFLU_RS14365	hypothetical protein
* NC_012660	445,413	0	G	A	39.1%	15.8 / 22.1	23	A128V (GCG→GTG)	PFLU_RS01980	primosomal protein N'
* NC_012660	5,247,207	0	C	A	38.5%	19.8 / 30.7	26	intergenic (-648/+4)	PFLU_RS23350/acs	ABC transporter substrate-binding protein/acetate--CoA ligase
* NC_012660	2,275,505	0	G	A	35.5%	45.0 / 11.7	32	E40E (GAG→GAA)	PFLU_RS10320	class I SAM-dependent methyltransferase
* NC_012660	4,304,394	0	C	A	35.3%	14.6 / 16.5	17	G198C (GGC→TGC)	PFLU_RS19025	fimbria/pilus outer membrane usher protein
* NC_012660	6,446,169	0	A	G	35.3%	15.3 / 12.7	17	E141E (GAA→GAG)	PFLU_RS28995	NADPH:quinone oxidoreductase family protein
* NC_012660	1,124,968	0	G	A	33.3%	17.0 / 14.4	18	T152I (ACC→ATC)	PFLU_RS05035	DUF1120 domain-containing protein
* NC_012660	6,241,996	0	G	A	33.3%	15.5 / 14.4	18	P92P (CCG→CCA)	PFLU_RS28025	NAD(P)/FAD-dependent oxidoreductase
* NC_012660	6,446,148	0	G	C	33.3%	19.2 / 16.2	19	R134R (CGG→CGC)	PFLU_RS28995	NADPH:quinone oxidoreductase family protein
* NC_012660	4,357,737	0	T	A	32.1%	46.5 / 30.5	28	K134M (AAG→ATG)	PFLU_RS19265	TetR family transcriptional regulator C-terminal domain-containing protein
* NC_012660	6,446,145	0	A	G	31.6%	24.0 / 12.3	19	Q133Q (CAA→CAG)	PFLU_RS28995	NADPH:quinone oxidoreductase family protein
* NC_012660	5,162,998	0	G	A	30.4%	60.9 / 37.6	46	A130V (GGG→GTG)	araG	L-arabinose ABC transporter ATP-binding protein AraG
* NC_012660	2,579,414	0	C	A	29.6%	39.7 / 21.2	27	V144F (GTC→ITC)	PFLU_RS11625	dihydroxy-acid dehydratase